EE1021 – Lecture 9 Home Worksheet

Topics: object-oriented programming (classes, instances, attributes, methods); special methods (`__init__`, `__str__`, comparisons, arithmetic); properties; class vs instance variables; composition.

Tip: Unless a task explicitly says "print", prefer returning values from methods. You can print returned values when testing.

Part A – Class Design & Methods

1. Battery – State and Behavior

Create a class `Battery` with the following behavior:

- Attributes:
  - `capacity_mAh`: total capacity in mAh.
  - `charge_mAh`: current charge in mAh (default `0`).
- Methods:
  - `charge(self, mAh)`:
    - Increases `charge_mAh` by `mAh`.
    - Never allow `charge_mAh` to exceed `capacity_mAh` (clamp to capacity).
    - Negative or zero values for `mAh` should raise `ValueError`.
  - `consume(self, current_mA, hours)`:
    - Reduces `charge_mAh` by `current_mA * hours`.
    - Never allow `charge_mAh` to go below `0` (clamp to `0`).
    - Negative or zero values for `current_mA` or `hours` should raise `ValueError`.
  - `level_pct(self)`:
    - Returns the current percentage as a `float` in `[0.0, 100.0]`.
  - `__str__(self)`:
    - Returns a string like `"Battery 76% (2280/3000 mAh)"`, where the percentage is rounded to the nearest integer.

2. Thermostat – Validation and Simple Logic

Create a class `Thermostat` with:

- Attributes:

  - `setpoint`: a `float` (target temperature in °C).
  - `mode`: a string in `{"heat", "cool", "off"}`.
  - `tolerance`: a `float` (default `0.5`, in °C).
- Methods:
  - `set_mode(self, mode)`:
    - Updates `mode`.
    - If `mode` is not one of `"heat"`, `"cool"`, or `"off"`, raise `ValueError`.
  - `needs_action(self, current_temp)`:
    - Returns:
      - `"heat_on"` if `mode == "heat"` and `current_temp < setpoint - tolerance`.
      - `"cool_on"` if `mode == "cool"` and `current_temp > setpoint + tolerance`.
      - `"idle"` otherwise.
  - `__str__(self)`:
    - Returns a string like `Thermostat(mode='heat', setpoint=22.0, tol=0.5)`.

Part B – Special Methods & Value Types

3. ColorRGB – String, Addition, Clamp

Create a class `ColorRGB(r, g, b)` representing an RGB color:

- Initialization:
  - Each channel (`r`, `g`, `b`) should be stored as an integer in `[0, 255]`.
  - If the provided value is outside `[0, 255]`, clamp it to that range (do **not** raise an error).
- Methods:
  - `to_hex(self)`:
    - Returns a string like `"#1a2b3c"`:
      - Lowercase hex.
      - Two hex digits per channel.
  - `__str__(self)`:
    - Returns the same value as `to_hex()`.
  - `__add__(self, other)`:
    - Returns a new `ColorRGB` where each channel is the average of `self` and `other`:
      - `r = round((self.r + other.r) / 2)` (similarly for `g`, `b`).
  - `lighter(self, percent)`:
    - Returns a new `ColorRGB` that is lightened by `percent`.
    - For example, `percent = 20` means "move 20% of the way from the current value toward `255`" for each channel.
    - Clamp each result to `255`.
  - `__eq__(self, other)`:
    - Two colors are equal if all three channels are equal.

4. TimeSpan – Normalization and Operators

Create a class `TimeSpan(hours=0, minutes=0, seconds=0)` representing a non-negative duration:

- Internals:
  - Store the duration in a normalized, non-negative form (e.g., only total seconds, or normalized `h:m:s`).
  - For example, `TimeSpan(minutes=90)` should be equivalent to `TimeSpan(hours=1, minutes=30)`.
- Methods:
  - `total_seconds(self)`:
    - Returns the total duration as an `int` (number of seconds).
  - `__add__(self, other)`:
    - Returns a new `TimeSpan` equal to the sum of the two durations.
  - `__sub__(self, other)`:
    - Returns a new `TimeSpan` equal to the difference.
    - If the result would be negative, clamp to zero (i.e., return a zero `TimeSpan`).
  - Comparisons:
    - Implement `__eq__(self, other)` and `__lt__(self, other)` based on `total_seconds()`.
  - `__str__(self)`:
    - Returns a string in the format `"HH:MM:SS"` with zero-padded fields.
- Class method:
  - `@classmethod from_seconds(cls, n)`:
    - Builds and returns a `TimeSpan` from `n` seconds (non-negative integer).

Part C – Containers & Composition (No Inheritance)

5. SlidingWindow – Running Statistics

Create a class `SlidingWindow(size)` that keeps the most recent values:

- Initialization:
  - `size` is the maximum number of values to remember.
  - If `size <= 0`, raise `ValueError`.
- Behavior:
  - Internally store up to `size` numeric values in order (oldest first).
- Methods:
  - `add(self, x)`:
    - Appends a new value `x`.
    - If the window would exceed `size`, drop the oldest value.
  - `mean(self)`:
    - Returns the average of the current values.
    - If the window is empty, raise `ValueError`.

  - `min(self)` and `max(self)`:
    * Return the minimum / maximum of the stored values.
    * If the window is empty, raise `ValueError`.
  - `__len__(self)`:
    * Returns the number of stored values.
  - `__str__(self)`:
    * Returns a compact summary like `"SlidingWindow(n=3, values=[1.0, 2.0, 3.0])"`.

6. Library – Tracking Loans

You will use composition (no inheritance) to model a simple library.

- Class `Member`:
  - `Member(member_id, name)`:
    * Stores:
      · `member_id` (an integer).
      · `name` (a string).
      · A collection of borrowed book titles (strings).
  - Methods:
    * `borrow(self, title)`:
      · Adds `title` to the member's borrowed collection.
      · Ignore duplicates (a title should appear at most once per member).
    * `return_book(self, title)`:
      · Removes `title` if it is currently borrowed.
      · If it is not present, do nothing.
    * `borrowed_count(self)`:
      · Returns the number of borrowed books.
- Class `Library`:
  - `Library()`:
    * Tracks members (you may store them in a list or dictionary).
    * Maintains an internal counter used to assign each new member a unique, incremental ID.
    * Store this counter as an **instance attribute**, not a class attribute.
  - Methods:
    * `enroll(self, name)`:
      · Creates a new `Member` with the next available ID.
      · Stores the member inside the `Library`.
      · Returns the new `Member` instance.
    * `find_by_id(self, member_id)`:
      · Returns the `Member` with the given `member_id`, or `None` if not found.
    * `total_borrowed(self)`:
      · Returns the total number of borrowed books across all mem-

bers.

Deliverables

- Submit a single `.py` file containing:
  - All the classes defined above.
  - Minimal test code (a few simple examples) that demonstrates each class works.
- You may also use the provided template: `Lecture9/EE1021_Lecture9_HomeWorksheet_Template.py`.
- Ensure that invalid inputs raise `ValueError` where specified, and that your code runs without errors.